



**VietNam National University
University of Engineering and Technology**

EMBEDDED SYSTEM FUNDAMENTALS
(ELT3240, NHẬP MÔN HỆ THỐNG NHÚNG)

Instructor: Dr. Nguyễn Kiêm Hùng

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Laboratory for Smart Integrated Systems



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COURSE OVERVIEW

Laboratory for Smart Integrated Systems

General Information

- **Instructor**

- Hung Nguyen-Kiem, PhD
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Objectives

- Put a large emphasis on architecture of embedded computing systems
 - Emphasis on ARM embedded processor micro-architectures (via the instruction set architecture) and ARM-based SoCs
 - Emphasis on the process and techniques for programming and interfacing embedded systems with real-world devices, embedded EDA tool chains, and application design
 - Demonstrating through practical examples
- Homework and a course project used to help you put the ideas from the course into practice

Objectives

After completing this course, you will master:

- Essential knowledge on embedded systems,
- Architecture of ARM-based embedded SoCs,
- Techniques for developing software components for ARM-based embedded SoCs,
- Using a typical toolchain to implement and test ARM-based embedded applications in C and assembly language

Contents

- **Topic 1: Introduction to Embedded Computing System**
 - What is embedded computing system?
 - Applications of embedded computing system,
 - ARM Development Environment,
 - C Language for Embedded System

Contents

- **Topic 2: Hardware architecture**
 - A brief of computer architecture taxonomy
Architecture of an embedded system: CPU, buses, I/O devices, memory and interfaces, ...
 - ARM processor and assembly language
 - ARM-Based Embedded System-on-Chips (SoCs)

Contents

- **Topic 3: Embedded Software**
 - The processes for compiling, assembling, and linking software components,
 - Developing some embedded software components
 - Basic concepts of processes and process-based design: Schedule and Inter-process communication,
 - Real-time operating systems (RTOS)

Contents

- **Topic 4: Interfacing to ARM Cortex-M**
 - I/O Programming based on GPIO ports
 - Interfacing and programming of serial UART ports
 - ADC and DAC and sensor interfacing
 - The Interrupt programming of the ARM
 - ...

Contents

- **Topic 5: Practices and Projects**
 - Labs: Write programs for a range of typical microcontroller peripherals (e.g. GPIO, USART, ADC, etc...) of **FRDM-KL46Z**
 - Assignment 1&2

Timetable

Week	Lecture Topic	Laboratory Work
1	Introduction to Embedded Systems and ARM Development Environment	Lab 1: MDK-ARM and FRDM-KL46Z: Getting Started
2	C Programming 1	Lab 2: Hello World
3	C Programming 2	Lab 3: Illustrating the use of C
4	ARM architecture and Assembly Language Programming	Lab 3: Assembly Language
5	ARM architecture and Assembly Language Programming	Lab4: SysTick Timer
6	ARM-based embedded System-on-Chips	Lab5: PIT, TPM Timer
7	Embedded Software	Assignment 1: Seat belt controller
8	Seminar 1	Assignment 1:
9	I/O Programming: GPIO	Lab 5: Controlling LED and SW
10	Interrupts	Lab 6: Interrupts and Timers
11	A-D /D-A and Sensor Interfacing	Lab 7: Visible Light Sensor
12	Embedded Communication Peripherals	Lab 8: UART
13	Real time Operating System	Lab 9: Keil RTX-RTOS
14	Review of ELT3240	Assignment 2: Final Project
15	Seminar 2	Assignment 2
ELT3240 Timetable v1.0		
Dr. Hung NK, 30.08.19		

Prerequisites

- Basic computer programming knowledge (e.g. C , Assembly language)
- Digital design knowledge and experience
- Basic knowledge of processor

Resources

- ARM
 - EDA/CAD tools: **Keil MDK-ARM IDE (v5)**
 - <https://www.keil.com/download/product/>
 - Evaluation Kit: **Freescale FRDM-KL46Z board**
 - Others: **Documents, AppNote, datasheet, Examples,...**
 - **My blog:** <https://dayhocstem.com/blog/dao-tao/he-thong-nhung>
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Teaching materials

- **Textbook(s):**

- Alexander G Dean: “Embedded Systems Fundamentals with ARM Cortex-M based Microcontrollers: A Practical Approach”, ARM Education Media (2017).
- Yifeng Zhu: “Embedded Systems with ARM Cortex-M Microcontrollers in Assembly Language and C: Third Edition”, [E-Man Press LLC](#) (2017).

- **Reference books/materials:**

- Trevor Martin: “The Designer’s Guide to the Cortex-M Processor Family”, Elsevier Ltd (2013).
- Joseph Yiu: “The Definitive Guide to the ARM Cortex-M0”, Elsevier Ltd (2011)

Administrative Details

- Marks
 - Mid-term exam (Assignments): 40%
 - Final Exam (writing/Oral): 60%
- Students have to be present:
 - at least 80% of the course meetings

Lectures, Homework, Quizzes

- It is important to attend each lecture
 - Lots of the material covered in lecture is not in the book or readings
- There will be several homework assignments
 - Not all will be collected/graded
 - May require use of C/C++ and Assembly
- There will be in-class quizzes

Project [1]

- For the course project, you will be expected to complete original work related to embedded systems in teams of **two** or **three** students.
- Projects will consist of **source code**, **final report**, and short presentation.
- You are encouraged to come up with your own topic for the project, but will be provided with a list of possible ideas.

Project [2]

- The course project counts for 40% of the overall grade.
- Students will be expected to turn in a summary of the contribution to the project to facilitate fair grading to individual team members.
- Additional details on the course projects will be provided in a separate document.

Honor

- You are encouraged to collaborate with other students in projects
- Final practice and project report, homework should be done by yourself
- Final Exam is closed book, closed notes... (only pen, blank paper are allowed)



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Thank You

Questions?

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